Adolescent substance use patterns and subsequent risk of mental and behavioural disorders, substance use, and suicidal behaviour: a cohort study

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Summary

Background **Despite growing knowledge of distinct patterns of substance use in adolescents, their associations with later adverse mental health and substance use outcomes remain poorly understood. We aimed to identify adolescent substance use patterns, describe adolescents with distinct patterns, and test the hypothesis that substance use patterns predict mental disorders, hospital contacts due to alcohol or drug use, and suicidal behaviour.**

Methods **We used a school-based survey of 68 301 adolescents (mean age 17·8 [SD 0·99]) in upper-secondary education, to identify substance use patterns using latent class analysis. Adolescents with distinct patterns were characterised by individual, social, and environmental health-related factors using logistic regression. First-time hospital contacts attributable to alcohol use, drug use, any mental or behavioural disorder, suicidal behaviour, eating disorder, anxiety or obsessive-compulsive disorder, and depression were assessed using hospital and prescription register data, with follow-up occurring from 2014 to 2022. Associations between substance use patterns and outcomes were examined using Cox regression.**

Findings **We identified the substance use patterns alcohol only (48·8%), frequent binge drinking (23·3%), experimental substance use (16·3%), and early-onset multiple substance use (11·6%). Adverse childhood experiences, economic disadvantages, and poor physical and mental health were accumulated in the experimental substance use and the early-onset multiple substance use groups. Notably, the risks of most studied outcomes were consistently higher in the experimental substance use and early-onset multiple substance use groups. Compared with the alcohol only group, the early-onset multiple substance use group had the highest risks, including alcohol-attributable hospital contacts (hazard ratio 2·25 [95% CI 1·89–2·68]), drug-attributable hospital contacts (10·7 [8·78–13·1]), any mental or behavioural disorder or use of psychotropic medication (1·58 [1·48–1·69]), and suicidal behaviour (3·20 [2·55–4·03]). Surprisingly, the frequent binge drinking group had a lower risk of any mental or behavioural disorder (0·83 [0·77–0·89) than the alcohol only group. The higher risk of all outcomes in the experimental and early-onset multiple substance use groups remained across the 8 years of follow-up.**

Interpretation **Adolescents with high-risk substance use face an increased risk of adverse mental and substance use outcomes in young adulthood, and health and social disadvantages tend to cluster within this group. Targeted preventive efforts should prioritise these vulnerable young people.**

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Introduction

Mental health issues often emerge during adolescence, with half of all lifetime mental and behavioural disorders developing by the mid-teens, and three-quarters by the mid-20s.[1] Adolescence is also a period when many young people begin experimenting with substance use. Among European students aged 15–16 years, nearly 80% have tried drinking alcohol, 41% have smoked cigarettes, and 17% have used illicit drugs at least once.[2] Danish adolescents report even higher rates than the European average, particularly for binge drinking.[2] In Denmark, alcohol can be legally purchased from the age of 16 years and is easily accessible in shops 24 h a day. The age limit for buying cigarettes is 18 years, but enforcement of both age restrictions is inconsistent.

Most young people drink for social or enhancement motives, seeking enjoyment.[3] Still, for some adolescents, substance use serves as a coping mechanism for dealing with familial challenges, peer pressure, loneliness, psychiatric symptoms, or school-related problems.[4,5] Moreover, social determinants of health, such as exposure to adverse childhood experiences or familial socioeconomic disadvantages, heighten the risk of mental and behavioural disorders and substance use disorders during adolescence and into adulthood.[6–8] These observations highlight the crucial role of an individual's environment in shaping both substance

Research in context

Evidence before this study

We reviewed the literature in PubMed without language or time restrictions on Jan 7, 2025, using the search terms: (adolescents OR youth OR young adults) AND (latent class analysis OR LCA) AND (substance use) AND (mental disorder OR psychiatric disorder OR depression OR anxiety OR eating disorder OR obsessive-compulsive disorder OR self-harm OR suicide attempt). Previous studies have identified distinct substance use patterns in adolescents, but only a few have explored their links to mental health outcomes. Current evidence suggests that high-risk substance use in adolescence is associated with various mental health issues. However, most existing studies were cross-sectional, relied on self-reported mental health symptoms, or were based on small samples, limiting their generalisability. There is a recognised need for further research into prospective associations between substance use patterns and mental health issues in larger scale studies.

Added value of this study

We identified distinct substance use patterns using data from a large cohort of individuals aged 15–19 years in Denmark. Concerningly, adolescents with most substance use were also burdened by a pronounced clustering of social and familial disadvantages, such as parental alcohol problems or economic hardship. These challenges were far less common among those with little or no substance use. Compared with adolescents with little substance use, those with the most substance use were much more likely to require hospital care for alcohol-related and drug-related issues, develop serious mental disorders such as depression or anxiety, or engage in suicidal behaviour. Our study shows that adolescent substance use can predict both immediate and long-term adverse outcomes.

Implications of all the available evidence

Evidence shows that adolescent substance use is not a uniform phenomenon but unfolds along distinct pathways with differing health implications. Preventive strategies should prioritise adolescents exhibiting high-risk substance use patterns by implementing evidence-based policy measures, such as raising the minimum legal drinking age, increasing alcohol prices, and restricting alcohol advertising. In addition, targeted initiatives should address the social-related and health-related disadvantages these vulnerable adolescents face. A comprehensive approach integrating social, familial, and health interventions will be essential to support adolescent wellbeing.

use and mental health, with these factors connected in a complex manner.

The high prevalence of alcohol and other substance use among young people is concerning, as the adolescent brain is particularly susceptible to harmful effects.[9] Alcohol, cannabis, and other drug use during adolescence have each been individually linked to an increased risk of later mental and substance abuse problems. For instance, frequent alcohol consumption or early drinking onset has been associated with an increased risk of depression and substance-use disorders later in life.[10,11] Similarly, cannabis use has been linked to suicidal ideation,[12] and both cannabis and other drug use during this stage of life have been associated with later depression and substance-use disorders.[11]

Furthermore, many adolescents engage in the use of multiple substances, often following distinct patterns.[13] Evidence suggests that adolescent substance use can be categorised into subgroups, with the largest proportion typically falling into a low-use category, mainly characterised by low levels of alcohol consumption. High multi-substance use groups tend to be the smallest and are characterised by high use of alcohol, tobacco, cannabis, and other illicit drugs.[13] Despite this knowledge, little is known about how specific substance use patterns predict adolescents' risk of developing mental and behavioural disorders and what distinguishes adolescents who engage in different patterns. Most existing studies are cross-sectional, rely on self-reported mental health symptoms, or are based on small samples, limiting their generalisability.[14–16] Addressing this gap is essential for identifying adolescents at the highest risk and determining whether these substance use patterns only reflect temporary issues or can serve as early indicators of future mental health concerns.

We aimed to assess whether adolescents' substance use patterns predict mental disorders and hospitalisation due to substance use. Our analyses specifically focused on suicidal behaviour, eating disorders, anxiety and obsessive-compulsive disorder (OCD), and depression, as these conditions place a substantial burden on adolescent health. The prevalence of eating disorders has risen in recent decades, and anxiety and depressive disorders rank among the leading causes of years lived with disability among mental health conditions.[17] We hypothesised that adolescents with the most substance use would exhibit greater disadvantages in baseline characteristics and face a higher risk of adverse outcomes over time.

Methods

Overview

We conducted a prospective cohort study linking survey data from 68 301 (40 807 females, 27 494 males) adolescents aged 15–19 years, collected in 2014 through the Danish National Youth Study (DNYS),[18] to nationwide health and demographic registers. Survey completion marked inclusion, and participants were followed up for approximately 8 years (2014–22). Personal identification numbers enabled individual and parental linkage to registers. We used parental linkage to obtain information on parental mental disorders, separation, education, and income to characterise adolescents with distinct substance use patterns at baseline.

Data collection for the DNYS 2014 was approved by the Danish Data Protection Agency (2013–54–0526). Participants provided informed consent before completing the DNYS 2014 online and, by participating, they consented to the use of their data for research purposes. Ethics approval for the current study was not required under Danish law, as there is no formal agency for approving register-based studies in Denmark. Nevertheless, the study follows the highest ethical research standards.

Data sources

The DNYS 2014 is a nationwide survey completed by 75 853 students from high schools and vocational schools and is considered representative of Danish upper-secondary students.[18] In Denmark, high schools prepare students for higher education, whereas vocational schools offer training for specific skilled professions. Data were collected during school hours and covered various topics related to health and wellbeing.[18]

All 137 Danish high schools were invited to participate, and 119 (87%) agreed, resulting in 70 674 students responding (85% of those invited). Additionally, 12 of the largest Danish vocational schools were invited, and 10 agreed (83%), resulting in 5179 students responding (69% of those invited). In this study, we focused on adolescents, which is typically defined as ages 10–19 years, with early adolescence spanning ages 10–14 years and late adolescence ages 15–19 years.[19] In Denmark, most students enter upper-secondary education at age 15 years and very few were younger than this in our dataset (<50). Therefore, we excluded participants outside the late adolescence age range (n=3762). Additionally, we excluded 3597 individuals due to missing identification numbers and 193 due to incomplete baseline data. The final sample comprised 68 301 students (appendix p 12). The sample was 40 807 (59·7%) females and 27 494 (40·3%) males, reflecting gender distributions in Danish upper-secondary education.[18]

Data on first-time hospital contacts were obtained from the Danish National Patient Register, which includes records of all contacts with public and private hospitals, including emergency departments, psychiatric units, and outpatient services.[20] The primary clinical condition is recorded as the action diagnosis (A-diagnosis), which represents the main reason for hospitalisation. Secondary conditions are recorded as B-diagnoses, which supplement the A-diagnosis by, for example, describing underlying diseases related to the current patient contact.[21]

Information on redemption of prescription medications was retrieved from the Danish National Prescription Register, which contains data on all prescription medications redeemed at Danish pharmacies.[22]

Exposures

We used latent class analysis (LCA) to identify subgroups with distinct substance use patterns based on 14 indicator variables derived from the DNYS. The indicators covered alcohol use (typical weekly intake, binge drinking, and early drunkenness), tobacco use (daily and occasional smoking and early onset), use of other tobacco-related and nicotine-related products, and illicit drug use (having tried cannabis, cannabis use in the past 30 days, and having tried other illicit drugs). Most indicator variables did not specify a timeframe, except for two questions referring to the past 30 days. We recoded responses into indicators representing behaviour present versus behaviour not present, except weekly alcohol intake, which we categorised as 0, 1–9, and ≥10 drinks in a typical week (appendix p 3).

Outcomes

The outcomes were defined as incident cases and included fully alcohol-attributable hospital contacts, drug-attributable hospital contacts, any mental or behavioural disorder or use of psychotropic medication (excluding mental or behavioural disorders caused by alcohol and other substance use), any mental or behavioural disorder (excluding mental or behavioural disorders caused by alcohol and other substance use), suicidal behaviour, eating disorder, anxiety disorder or OCD, and depression. The complete list of codes used to define each outcome, along with written descriptions, is provided in the appendix (pp 2, 4–5).

Statistical analysis

We conducted descriptive statistics to characterise participants at baseline, stratified by sex, presented as frequencies and proportions. The LCA was conducted in SAS version 9.4, following Weller and colleagues.[23] We estimated models with one to seven classes separately for females and males. Model selection was based on Bayesian Information Criterion, Akaike Information Criterion, likelihood ratio χ^2 test, entropy, and the average latent class posterior probability (appendix p 8). The final class enumeration was guided by these criteria, class sizes, and theoretical support for the identified classes.[23] We selected a four-class model for both sexes. Individuals were assigned to the class for which they had the highest posterior membership probability.

See Online for appendix

Adolescents with distinct substance use patterns were characterised using logistic regression. Adjusted first-time incidence rates per 10 000 person-years were estimated using Poisson regression with an offset for risk-time. Cox regression models were used to estimate prospective associations, with follow-up from the survey date until the outcome, death, or end of follow-up (ending Dec 31, 2022). Participants with the outcome of interest within 4 years before baseline were excluded from analyses of that outcome to ensure incident cases. Proportional hazard assumptions were tested graphically and statistically. For

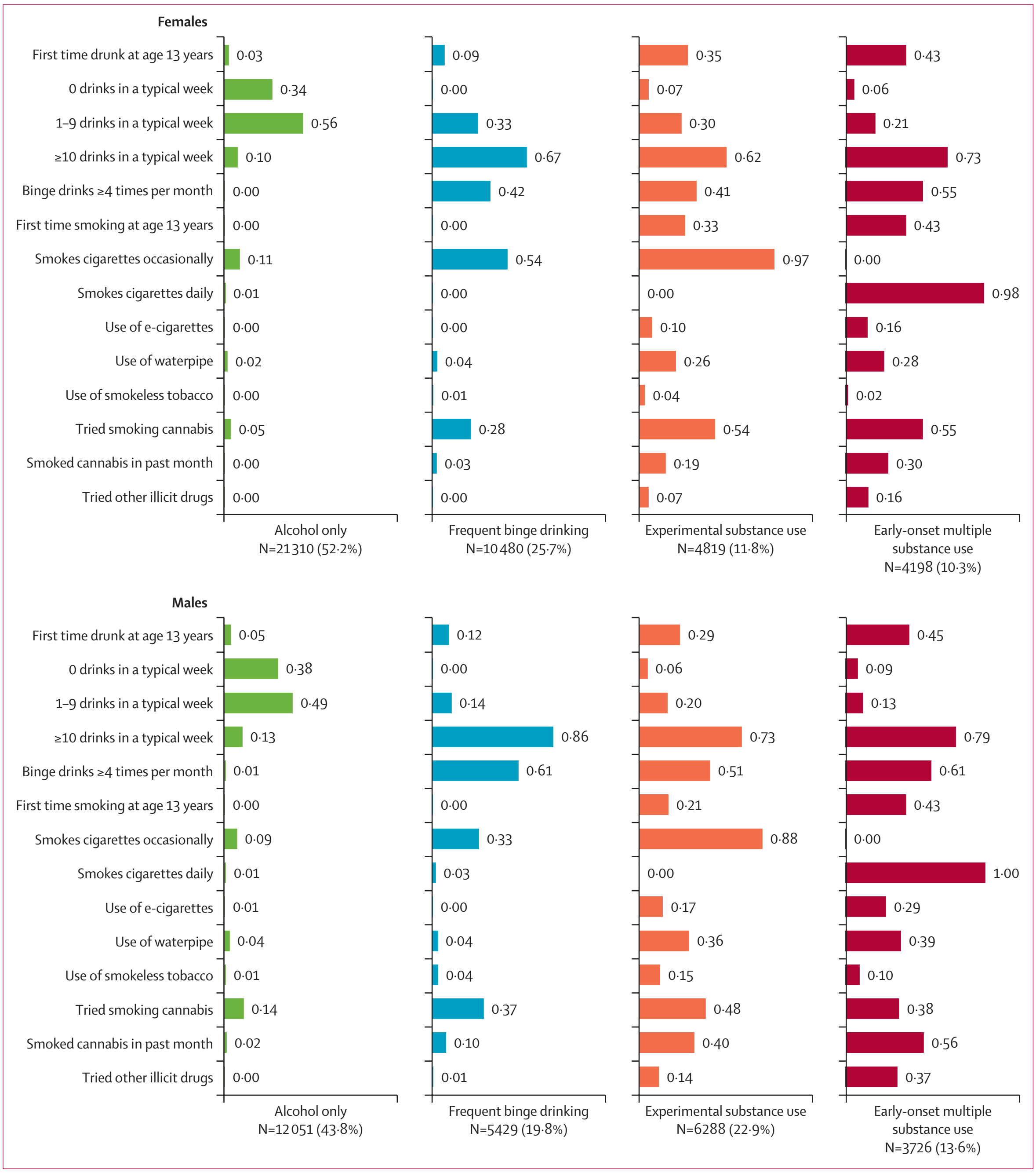

***Figure 1:* Bar chart illustrating the loading of individual substance use indicators in substance use subgroups, in females and males (15–19 years)**
Data are from the Danish National Youth Cohort 2014 (n=68 301). 40 807 (59·7%) females and 27 494 (40·3%) males.

the outcomes alcohol-attributable hospital contacts, mental and behavioural disorder or use of psychotropic medication, and suicidal behaviour, p values were significant, likely due to high statistical power, which allowed for the detection of even small differences. We stratified the follow-up time into two periods: 0·0–3·9 years and 4·0–8·2 years following baseline, to assess whether the effects remained stable over time. Complete data were available for all covariates, and robust standard errors were applied in all analyses to account for school-level clustering. Due to few eating disorder cases among males, those analyses were restricted to females.

Associations were first examined across the entire sample, testing for effect modification by sex. Additional models were stratified by follow-up time and subsequently by sex.

All models were adjusted for age (continuous), sex (male *vs* female), school year (first, second, or third),

	Alcohol only (33 361 [48·8%])		Frequent binge drinking (15 909 [23·3%])		Experimental substance use (11 107 [16·3%])		Early-onset multiple substance use (7924 [11·6%])	
	n (%)	OR (95% CI)	n (%)	OR (95% CI)	n (%)	OR (95% CI)	n (%)	OR (95% CI)
Adverse childhood experiences								
Comes from a separated family*	9108 (27·3%)	1·00 (ref)	4801 (30·2%)	1·15 (1·10–1·20)	4038 (36·4%)	1·57 (1·49–1·65)	3739 (47·2%)	2·27 (2·15–2·41)
Parental alcohol problems	1559 (4·7%)	1·00 (ref)	862 (5·4%)	1·14 (1·05–1·25)	928 (8·4%)	2·02 (1·84–2·21)	1035 (13·1%)	2·98 (2·73–3·25)
Parental mental disorder*	2202 (6·6%)	1·00 (ref)	902 (5·7%)	0·89 (0·82–0·97)	819 (7·4%)	1·16 (1·08–1·24)	814 (10·3%)	1·54 (1·41–1·67)
Death of a parent or sibling	1822 (5·5%)	1·00 (ref)	737 (4·6%)	0·93 (0·86–1·01)	651 (5·9%)	1·09 (1·00–1·19)	733 (9·3%)	1·44 (1·29–1·61)
Parental financial situation								
Parents have difficulties paying bills	4497 (13·5%)	1·00 (ref)	2160 (13·6%)	1·07 (1·00–1·13)	981 (17·8%)	1·61 (1·52–1·72)	1985 (25·1%)	2·29 (2·11–2·48)
Parents receive social benefits or transfer income*	5635 (16·9%)	1·00 (ref)	1652 (10·4%)	0·69 (0·65–0·75)	1654 (14·9%)	0·96 (0·90–1·02)	1658 (20·9%)	1·29 (1·20–1·39)
Low family income*	6636 (19·9%)	1·00 (ref)	1886 (11·9%)	0·66 (0·62–0·71)	1885 (17·0%)	0·92 (0·87–0·98)	1894 (23·9%)	1·22 (1·12–1·32)
One or two parents have a low education*	13 515 (40·5%)	1·00 (ref)	5676 (35·7%)	0·86 (0·81–0·90)	4181 (37·6%)	0·94 (0·88–1·00)	3515 (44·4%)	1·04 (0·94–1·15)
Physical health								
Frequent sleep problems	5433 (16·3%)	1·00 (ref)	2842 (17·9%)	1·16 (1·10–1·22)	2617 (23·6%)	1·80 (1·70–1·89)	2410 (30·4%)	2·51 (2·35–2·68)
Poor self-perceived health	2449 (7·3%)	1·00 (ref)	1019 (6·4%)	0·88 (0·81–0·95)	977 (8·8%)	1·37 (1·26–1·49)	1345 (17·0%)	2·69 (2·47–2·94)
Daily physical pain or discomfort	4355 (13·1%)	1·00 (ref)	2066 (13·0%)	0·99 (0·93–1·05)	1600 (14·4%)	1·41 (1·33–1·50)	1548 (19·5%)	1·82 (1·68–1·98)
Mental health								
Ever considered suicide	4501 (13·5%)	1·00 (ref)	1979 (12·4%)	0·91 (0·85–0·96)	1966 (17·7%)	1·56 (1·46–1·66)	1930 (24·4%)	2·15 (1·98–2·33)
Lead a stressful life	3444 (10·3%)	1·00 (ref)	1661 (10·4%)	1·03 (0·97–1·10)	1359 (12·2%)	1·50 (1·40–1·62)	1404 (17·7%)	2·20 (2·03–2·39)
Poor life satisfaction	2358 (7·1%)	1·00 (ref)	965 (6·1%)	0·85 (0·79–0·92)	917 (8·3%)	1·37 (1·27–1·49)	1029 (13·0%)	2·07 (1·90–2·27)
Peer and school relations								
Difficulties talking openly with friends	7097 (21·3%)	1·00 (ref)	2318 (14·6%)	0·63 (0·60–0·66)	1950 (17·6%)	0·78 (0·73–0·83)	1283 (16·2%)	0·71 (0·65–0·76)
Low school confidence	2052 (6·2%)	1·00 (ref)	1188 (7·5%)	1·19 (1·10–1·29)	1168 (10·5%)	1·95 (1·80–2·11)	1134 (14·3%)	2·71 (2·47–2·97)
Dislike going to school	4744 (14·2%)	1·00 (ref)	2337 (14·7%)	0·99 (0·93–1·04)	2186 (19·7%)	1·59 (1·50–1·68)	1933 (24·4%)	1·99 (1·85–2·15)
Feeling unsupported by teachers	3136 (9·4%)	1·00 (ref)	1709 (10·7%)	1·18 (1·11–1·25)	1585 (14·3%)	1·77 (1·66–1·89)	1285 (16·2%)	2·08 (1·92–2·25)

The Danish National Youth Cohort (n=68 301), Denmark, 2014. Original questions from the Danish National Youth Survey 2014 are listed in the appendix (p 6). ORs were adjusted for age, sex, school year, school type, and self-reported ethnicity. Standard errors were adjusted for clustering at school level. OR=odds ratio. *Information was obtained from registers.

Table 1: **Individual, social, and environmental health-related characteristics at baseline according to substance use pattern in individuals aged 15–19 years**

school type (vocational school, Higher General Examination Programme, or Higher Preparatory Examination Programme), and self-perceived ethnicity (Danish/Danish and other ethnicities/Non-Danish), measured at baseline.

Age and sex were obtained through linkage with the Danish Civil Registration System,[24] where sex typically reflects sex assigned at birth. School year, school type, and ethnicity were self-reported from the DNYS 2014. For ethnicity, multiple selections were permitted.

Role of the funding source

The funder of the study had no role in study design, data collection, data analysis, data interpretation, or writing the report.

Results

The study included 68 301 secondary school students aged 15–19 at baseline in 2014 (mean age 17·8 [SD 0·99]), with more female (40 807 [59·7%]) than male participants (27 494 [40·3%]; appendix p 7). Most participants attended high school, with a small proportion of females (567 [1·4%]) and males (2237 [8·1%]) attending vocational school. Most participants identified as Danish (61 362 [89·8%]), and 40% (27 347) were in their first year of upper-secondary education. Parental education was primarily short (26 887 [39·4%]) or medium length (25 958 [38·0%]). Most participants lived in suburbs (24 901 [36·5%]) or rural areas (23 528 [34·4%]). A small percentage (5866 [8·6%]) reported complete abstention from alcohol, tobacco, and drugs at baseline.

We identified four substance use patterns (figure 1). The patterns were named based on their characteristics: alcohol only (33 361/68 301 [48·8%]), frequent binge drinking (15 909/68 301 [23·3%]), experimental substance use (11 107/68 301 [16·3%]), and early-onset multiple substance use (7924/68 301 [11·6%]). For those with the alcohol only pattern, it was common to drink 0 (7245 [34%] of 21 310 females and 4579 [38%] of 12 051 males) or 1–9 (11 934 [56%] of 21 310 females and 5905 [49%] of 12 051 males) drinks in a typical week, with very low probabilities for any other substance use behaviours. For

	Cases (n/N)	Incidence rate (95% CI)*	HR (95% CI)*
Alcohol-attributable hospital contacts	**1185/67 666**		
Alcohol only	459/33 190	16·47 (14·66–18·27)	1·00 (ref)
Frequent binge drinking	265/15 765	20·14 (17·62–22·67)	1·22 (1·03–1·46)
Experimental substance use	220/10 964	24·72 (21·32–28·12)	1·50 (1·27–1·78)
Early-onset multiple substance use	241/7747	37·12 (31·02–42·23)	2·25 (1·89–2·68)
Drug-attributable hospital contacts†	**877/68 170**		
Alcohol only	154/33 346	5·30 (4·38–6·21)	1·00 (ref)
Frequent binge drinking	108/15 903	8·59 (7·02–10·1)	1·62 (1·30–2·02)
Experimental substance use	202/11 082	21·10 (17·90–24·31)	3·98 (3·30–4·81)
Early-onset multiple substance use	413/7839	56·80 (49·67–63·94)	10·71 (8·78–13·06)

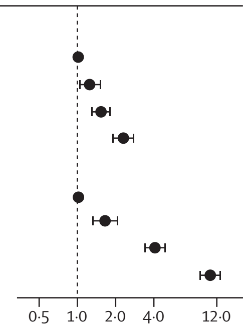

Figure 2: **Incidence rates and HRs of alcohol-attributable and drug-attributable hospital contacts by substance use patterns in individuals aged 15–19 years over 8 years of follow-up**

Data are from the Danish National Youth Cohort 2014 (n=68 301). Incidence rates are per 10 000 person-years. Standard errors were adjusted for clustering at school level. HR=hazard ratio. *Adjusted for age, sex, school year, school type, and self-reported ethnicity. †Substances other than alcohol.

those with the frequent binge drinking pattern, most consumed 10 or more drinks in a typical week (7021 [67%] of 10 480 females and 4669 [86%] of 5429 males). Frequent binge drinking was common (4402 [42%] of 10 480 females and 3312 [61%] of 5429 males), as was occasional cigarette smoking (5659 [54%] of 10 480 females and 1792 [33%] of 5429 males). For those with the experimental substance use pattern, besides drinking alcohol frequently, many occasionally smoked cigarettes (4674 [97%] of 4819 females and 5533 [88%] of 6288 males) and had tried smoking cannabis (2602 [54%] of 4819 females and 3018 [48%] of 6288 males). For those with the early-onset multiple substance use pattern, many had been drunk (1803 [43%] of 4198 females and 1677 [45%] of 3726 males) and smoked cigarettes (1805 [43%] of 4198 females and 1602 [43%] of 3726 males) before the age of 13 years. All individuals with the early-onset multiple substance use pattern smoked cigarettes daily, with high percentages having used cannabis in the past month (1259 [30%] of 4198 females and 2087 [56%] of 3726 males) and having tried other illicit drugs (672 [16%] of 4198 females and 1379 [37%] of 3726 males; figure 1). A description of the substance use patterns can also be found in the appendix (p 13).

The substance use patterns for females and males appeared to be similar, but the relative size of groups differed. 52·2% (21 310/40 807) of females versus 43·8% (12 051/27 494) of males belonged to the alcohol only group, and 11·8% (4819/40 807) of females versus 22·9% (6288/27 494) of males belonged to the experimental substance use group.

Individual, social, and environmental health-related characteristics at baseline varied across substance use patterns (table 1). Adolescents with the early-onset multiple substance use pattern consistently had the highest prevalence of the least favourable characteristics, followed by those with the experimental substance use pattern. Conversely, those with the frequent binge drinking pattern often had the lowest prevalence. For example, odds ratios (ORs) for ever having considered suicide were 0·91 (95% CI 0·85–0·96) in the frequent binge drinking group, 1·56 (1·46–1·66) in the experimental substance use group, and 2·15 (1·98–2·33) in the early-onset multiple substance use group, compared with the alcohol only group. Similar tendencies were observed for other baseline characteristics, including poor self-perceived health, poor life satisfaction, disliking going to school, having a parent with a mental disorder, and having parents who receive social benefits or transfer income. Having difficulties talking openly with friends was a notable characteristic, as all groups had ORs below 1 compared with the alcohol only group. The interaction of sex was tested statistically. Although some p values were significant, the direction and magnitude of the prevalence of individual, social, and environmental health-related characteristics were similar for females and males (appendix pp 9–10).

Adolescents in the early-onset multiple substance use group consistently exhibited the highest risk across all outcomes (figures 2, 3). For alcohol-attributable and drug-attributable hospital contacts, there was a gradient of higher risk across the substance use patterns (from frequent binge drinking to experimental substance use to early-onset multiple substance use) compared with adolescents in the alcohol only group (figure 2). During 548 153 person-years of follow-up, 1185 participants had an alcohol-attributable hospital contact. The hazard ratio (HR) for an alcohol-attributable hospital contact was 1·22 (1·03–1·46) in the frequent binge drinking group, 1·50 (1·27–1·78) in the experimental substance use group, and 2·25 (1·89–2·68) in the early-onset multiple substance use group, compared with the alcohol only group. Furthermore, adolescents in the early-onset multiple substance use group had an exceptionally high risk (10·7 [8·78–13·1]) for substance use-related hospital contacts compared with the alcohol only group.

The most common outcome was any mental or behavioural disorder or use of psychotropic medication, which occurred in 9425 of 64 586 adolescents during an average of 7·6 years of follow-up. The incidence rate was highest in the early-onset multiple substance use

	Cases (n/N)*	Incidence rate (95% CI)†	HR (95% CI)†
Any mental or behavioural disorder or use of psychotropic medication‡	**9425/64586**		
Alcohol only	4507/31686	175·27 (168·54–182·00)	1·00 (ref)
Frequent binge drinking	1890/15258	151·96 (144·06–159·86)	0·87 (0·82–0·91)
Experimental substance use	1520/10448	202·57 (190·44–212·70)	1·15 (1·09–1·22)
Early-onset multiple substance use	1508/7194	276·01 (259·63–292·39)	1·58 (1·48–1·69)
Any mental or behavioural disorder‡	**5666/65586**		
Alcohol only	2696/32132	98·40 (93·82–102·99)	1·00 (ref)
Frequent binge drinking	1069/15431	81·33 (75·40–87·27)	0·83 (0·77–0·89)
Experimental substance use	946/10626	120·35 (111·41–129·30)	1·22 (1·14–1·32)
Early-onset multiple substance use	955/7397	161·96 (150·0–174·14)	1·65 (1·52–1·79)
Suicidal behaviour	**542/67934**		
Alcohol only	207/33274	6·20 (5·24–7·16)	1·00 (ref)
Frequent binge drinking	81/15856	5·67 (4·42–6·93)	0·92 (0·71–1·19)
Experimental substance use	117/11027	13·24 (10·95–15·55)	2·14 (1·68–2·72)
Early-onset multiple substance use	137/7777	19·87 (16·41–23·32)	3·20 (2·55–4·03)
Eating disorder§	**699/40269**		
Alcohol only	343/21021	19·49 (17·22–21·79)	1·00 (ref)
Frequent binge drinking	164/10380	18·57 (15·48–21·65)	0·95 (0·79–1·15)
Experimental substance use	108/4741	27·15 (21·27–33·03)	1·39 (1·11–1·75)
Early-onset multiple substance use	84/4127	25·36 (19·91–30·81)	1·30 (1·03–1·64)
Anxiety disorder or OCD	**1825/67706**		
Alcohol only	915/33065	31·05 (28·73–33·37)	1·00 (ref)
Frequent binge drinking	321/15806	22·41 (19·72–25·10)	0·72 (0·63–0·83)
Experimental substance use	295/11015	34·39 (30·40–38·38)	1·11 (0·97–1·27)
Early-onset multiple substance use	294/7820	43·74 (38·29–49·18)	1·41 (1·22–1·63)
Depression	**2017/67782**		
Alcohol only	982/33154	32·78 (30·44–35·12)	1·00 (ref)
Frequent binge drinking	418/15820	29·67 (26·80–32·54)	0·90 (0·81–1·01)
Experimental substance use	299/11005	35·16 (30·62–39·70)	1·07 (0·93–1·24)
Early-onset multiple substance use	318/7803	48·76 (42·66–54·86)	1·49 (1·30–1·70)

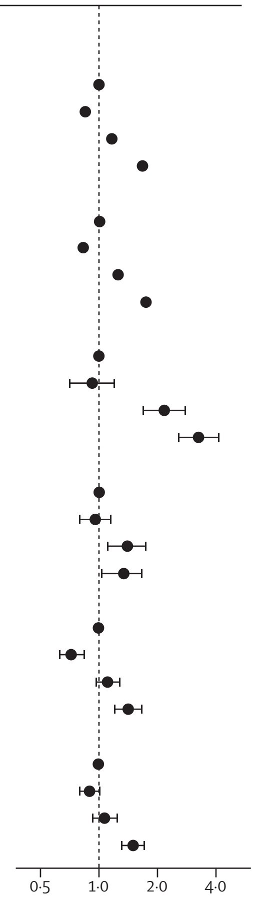

Figure 3: **Incidence rates and HRs of adverse mental health outcomes by substance use patterns in individuals aged 15–19 years over 8 years of follow-up**
Data are from the Danish National Youth Cohort 2014 (n=68 301). Incidence rates are per 10 000 person-years. Standard errors were adjusted for clustering at school level. HR=hazard ratio. OCD=obsessive-compulsive disorder. *Excludes participants who had the outcome of interest within 4 years before baseline. †Adjusted for age, sex, school year, school type, and self-reported ethnicity, ‡Excluding mental or behavioural disorders caused by alcohol or substance use. §Cases per person-years, incidence rates, and HRs exclusively among females.

group, at 276 (260–292) per 10 000 person-years, and lowest in the frequent binge drinking group, at 152 per 10 000 person-years (144–160). The HR for any mental or behavioural disorder or use of psychotropic medication was 0·87 (0·82–0·91) in the frequent binge drinking group, 1·15 (1·09–1·22) in the experimental substance use group, and 1·58 (1·48–1·69) in the early-onset multiple substance use group, compared with the alcohol only group.

During follow-up, there were 5666 cases of any mental or behavioural disorder, 542 cases of suicidal behaviour, 699 cases of eating disorder (only among females), 1825 cases of anxiety or OCD, and 2017 cases of depression. Compared with adolescents in the alcohol only group, adolescents in the frequent binge drinking group generally had a lower risk of these outcomes, whereas the experimental substance use and early-onset multiple substance use groups had a higher risk. For example, HRs for anxiety or OCD were 0·72 (0·63–0·83) in the frequent binge drinking group, 1·11 (0·97–1·27) in the experimental substance use group, and 1·41 (1·22–1·63) in the early-onset multiple substance use group, compared with the alcohol only group. Notably, the early-onset multiple substance use group showed a very high risk of suicidal behaviour (3·20 [2·55–4·03]) compared with the alcohol only group.

Results appeared similar in strata of sex (appendix p 11). Consistent with this finding, we did not observe any statistically significant modification of the effect of substance use pattern on the outcomes of interest by sex (all p values >0·05), except for the interaction with any mental or behavioural disorder, where the p value was 0·0004.

When splitting the follow-up time into two strata, nearly all associations persisted with similar magnitudes across both periods (table 2). For example, HRs for any mental or behavioural disorder or use of psychotropic medication were 0·89 (0·83–0·96) in the frequent binge

	0·0–3·9 years' follow-up		4·0–8·2 years' follow-up	
	Cases	HR (95% CI)	Cases	HR (95% CI)
Alcohol-attributable hospital contacts				
Alcohol only	329	1·00 (ref)	130	1·00 (ref)
Frequent binge drinking	195	1·26 (1·03–1·55)	70	1·12 (0·85–1·49)
Experimental substance use	149	1·42 (1·15–1·76)	71	1·69 (1·29–2·23)
Early-onset multiple substance use	143	1·85 (1·48–2·31)	98	3·27 (2·50–4·29)
Drug-attributable hospital contacts*				
Alcohol only	72	1·00 (ref)	82	1·00 (ref)
Frequent binge drinking	60	1·92 (1·39–2·64)	48	1·36 (0·98–1·90)
Experimental substance use	108	4·50 (3·35–6·03)	94	3·53 (2·62–4·77)
Early onset multiple substance use	230	12·67 (9·50–16·90)	183	8·98 (6·98–11·54)
Any mental or behavioural disorder or use of psychotropic medication†				
Alcohol only	1951	1·00 (ref)	2556	1·00 (ref)
Frequent binge drinking	834	0·89 (0·83–0·96)	1056	0·85 (0·79–0·92)
Experimental substance use	633	1·12 (1·03–1·22)	887	1·18 (1·09–1·27)
Early-onset multiple substance use	717	1·67 (1·52–1·84)	791	1·51 (1·37–1·66)
Any mental or behavioural disorder†				
Alcohol only	1339	1·00 (ref)	1357	1·00 (ref)
Frequent binge drinking	527	0·83 (0·75–0·92)	542	0·82 (0·74–0·92)
Experimental substance use	445	1·18 (1·06–1·30)	501	1·27 (1·15–1·40)
Early-onset multiple substance use	494	1·67 (1·47–1·89)	461	1·62 (1·42–1·85)
Suicidal behaviour				
Alcohol only	108	1·00 (ref)	99	1·00 (ref)
Frequent binge drinking	51	1·13 (0·80–1·61)	30	0·69 (0·45–1·04)
Experimental substance use	68	2·48 (1·78–3·46)	49	1·76 (1·23–2·52)
Early-onset multiple substance use	93	4·35 (3·22–5·87)	44	2·02 (1·42–2·88)
Eating disorder‡				
Alcohol only	199	1·00 (ref)	144	1·00 (ref)
Frequent binge drinking	96	0·96 (0·73–1·27)	68	0·94 (0·70–1·26)
Experimental substance use	56	1·24 (0·90–1·72)	52	1·60 (1·16–2·21)
Early-onset multiple substance use	47	1·24 (0·89–1·73)	37	1·38 (0·96–1·99)
Anxiety or OCD				
Alcohol only	472	1·00 (ref)	443	1·00 (ref)
Frequent binge drinking	159	0·70 (0·58–0·84)	162	0·75 (0·62–0·90)
Experimental substance use	141	1·04 (0·85–1·28)	154	1·18 (0·98–1·41)
Early-onset multiple substance use	151	1·38 (1·14–1·68)	143	1·43 (1·16–1·78)
Depression				
Alcohol only	436	1·00 (ref)	546	1·00 (ref)
Frequent binge drinking	202	0·97 (0·84–1·12)	216	0·85 (0·73–1·00)
Experimental substance use	130	1·05 (0·86–1·28)	169	1·09 (0·90–1·31)
Early-onset multiple substance use	145	1·51 (1·25–1·84)	173	1·47 (1·24–1·74)

Danish National Youth Cohort (n=68 301), Denmark, 2014. HRs were adjusted for age, sex, school year, school type, and self-reported ethnicity. Standard errors were adjusted for clustering at school level. *Substances other than alcohol. †Excluding mental or behavioural disorders caused by alcohol or substance use, ‡Cases and HRs exclusively among females. HR=hazard ratio. OCD=obsessive-compulsive disorder.

Table 2: **HRs of alcohol and substance use-related hospital contacts and adverse mental health outcomes by substance use patterns in individuals aged 15–19 years within 3·9 and 8·2 years of follow-up**

drinking group, 1·12 (1·03–1·14) in the experimental substance use group, and 1·67 (1·52–1·84) in the early-onset multiple substance use groups 0·0–3·9 years following baseline. In comparison, HRs for the same groups in the subsequent period (4·0–8·2 years following baseline) were 0·85 (0·79–0·92), 1·18 (1·09–1·27), and 1·51 (1·37–1·66).

Discussion

In this large cohort of 68 301 adolescents, we identified four distinct substance use patterns: alcohol only, frequent binge drinking, experimental substance use, and early-onset multiple substance use. In those with most substance use (experimental substance use and early-onset multiple substance use), we observed a pronounced clustering of social and familial disadvantages and poor mental and physical health when compared with those with little substance use (alcohol only and frequent binge drinking). Interestingly, adolescents with the frequent binge drinking pattern stood out as they were least likely to report these disadvantages. The substance use patterns predicted future adverse outcomes. For alcohol-attributable and drug-attributable hospital contacts, associations reflected a dose–response relationship across the four groups. Consistently, adolescents with the most substance use had the highest risks across all outcomes. Notably, adolescents with the frequent binge drinking pattern had the lowest risks of mental and behavioural disorders. Results were similar when follow-up time was split into 0·0–3·9 years and 4·0–8·2 years following baseline, showing the predictive value of adolescent substance use into early adulthood.

Our findings align with previous research that identified similar substance use patterns among adolescents, including both low-use and high multi-use patterns.[13,25] The replication of these patterns across studies reinforces their validity. We further show that these patterns are strong predictors of short-term and long-term adverse outcomes.

Several mechanisms might explain our findings. Risky substance use naturally increases the likelihood of situations leading to alcohol-related or drug-related hospital contacts. Additionally, since high alcohol consumption in adolescence often persists into adulthood and is linked to later alcohol use disorder,[26] adolescents with high-risk use at baseline might remain exposed to harm over time. The relationship between substance use and mental disorders is more complex and probably bidirectional: individuals might use substances to cope with emotional symptoms,[27] and substance use can worsen mental health problems,[28] creating a self-reinforcing cycle. Broader social determinants probably also have a role, as chronic stress from adverse conditions can drive both mental health problems and substance use.[29] This complex relationship aligns with our finding that those with the riskiest use patterns were also the most disadvantaged and reported the poorest mental health at baseline. Interestingly, adolescents with the frequent binge drinking pattern had a lower risk of developing mental disorders than those with the least substance use (alcohol only). In Denmark, drinking alcohol, particularly binge drinking, is common[2] and socially embedded, often reflecting social participation,[30] which might support and reflect mental wellbeing. Conversely, adolescents who drink less or not at all might be more socially excluded. This hypothesis is supported by our baseline data showing that the alcohol only group had the most difficulty talking openly with their friends compared with the other groups. Additionally, adolescents with the frequent binge drinking pattern had the most favourable socioeconomic conditions at baseline, an established protective factor against mental disorders.[31]

Strengths of this study include its large sample of adolescents and the long follow-up period using national registers, reducing selection bias and minimising attrition. Although substance use data relied on self-reports, which might vary in validity depending on socioeconomic position,[32] the patterns we identified were supported by the dose–response relationship we observed with alcohol-attributable and drug-attributable hospital contacts. Some substance use indicators were lifetime measures without specific time frames, limiting our ability to investigate whether individuals progress from one type of substance use pattern to another.

Limitations include the reliance on hospital contacts for outcomes, which probably underestimates milder cases of mental disorders and suicidal behaviours. Although we included psychotropic medication use and probable self-harm to broaden the definitions, some cases might have been misclassified. Finally, pre-hospital contacts were not considered, probably underestimating the actual burden of alcohol-related harm, as many cases do not require hospitalisation.[33] Thus, although our study is strong in capturing severe cases, it probably under-represents less acute cases, reflecting only the tip of the iceberg. We also did not disaggregate results by ethnicity, so potential differences in substance use patterns or outcomes across ethnic groups will not have been fully captured. In Denmark, hospital-based health care is free, and young people generally begin drinking from an early age.[2] Thus, findings from our study might not be directly generalisable to countries with markedly different health-care systems or social norms regarding alcohol consumption.

A substantial proportion of adolescents engaged in frequent binge drinking or multiple substance use, challenging the perception that such behaviours are limited to a small, atypical subset. Those who mainly binge drank appeared well functioning at baseline and had a lower risk of developing mental disorders than all other groups in the study, but their high alcohol consumption still raises concerns, as it might persist into adulthood and pose other long-term health risks or lead to more harmful substance use behaviours. Alcohol use in adolescence might act as a gateway to the use of other substances,[9,34] highlighting the need for prevention even among seemingly well functioning adolescents. More longitudinal data on adolescent substance use are needed to better understand how substance use patterns evolve from adolescence into young adulthood. In contrast, those with riskier substance use patterns faced higher risks of developing mental disorders, highlighting the need for targeted support. Proven policy measures, such as increasing the minimum legal drinking age, increasing alcohol prices, and restricting advertising, could help protect these disadvantaged adolescents.[35] Overall, our study shows that adolescent substance use patterns are linked to both immediate and long-term adverse outcomes. Recognising these patterns could help guide future prevention efforts in a diverse group of young people.

Contributors

JST planned and designed the study. JST and ERH analysed the data. ERH wrote the first draft of the manuscript. JST and SK verified the underlying data. All authors contributed to the interpretation of study results, critical revision of the paper, and approval of the final version, and agree to be accountable for all aspects of this Article.

Declaration of interests

We declare no competing interests.

Data sharing

Data from the Danish National Youth Study can be shared after reasonable request to the corresponding author.

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